

The Power of Vaccination

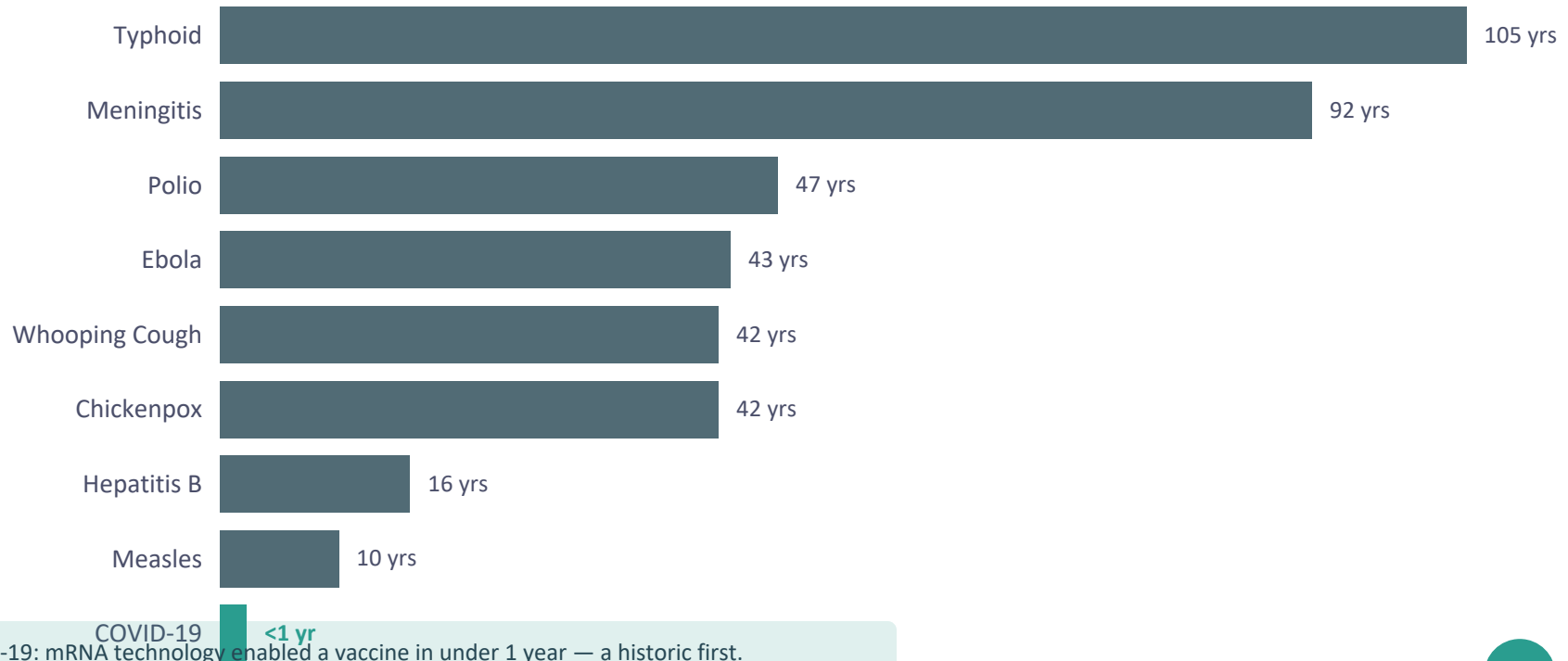
From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview spanning 1796–2020

Source: Our World in Data

From 100 Years to Under 1 Year

Vaccine Development Is Accelerating



COVID-19: mRNA technology enabled a vaccine in under 1 year — a historic first.

Source: Our World in Data | Roush & Murphy, 2007

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox

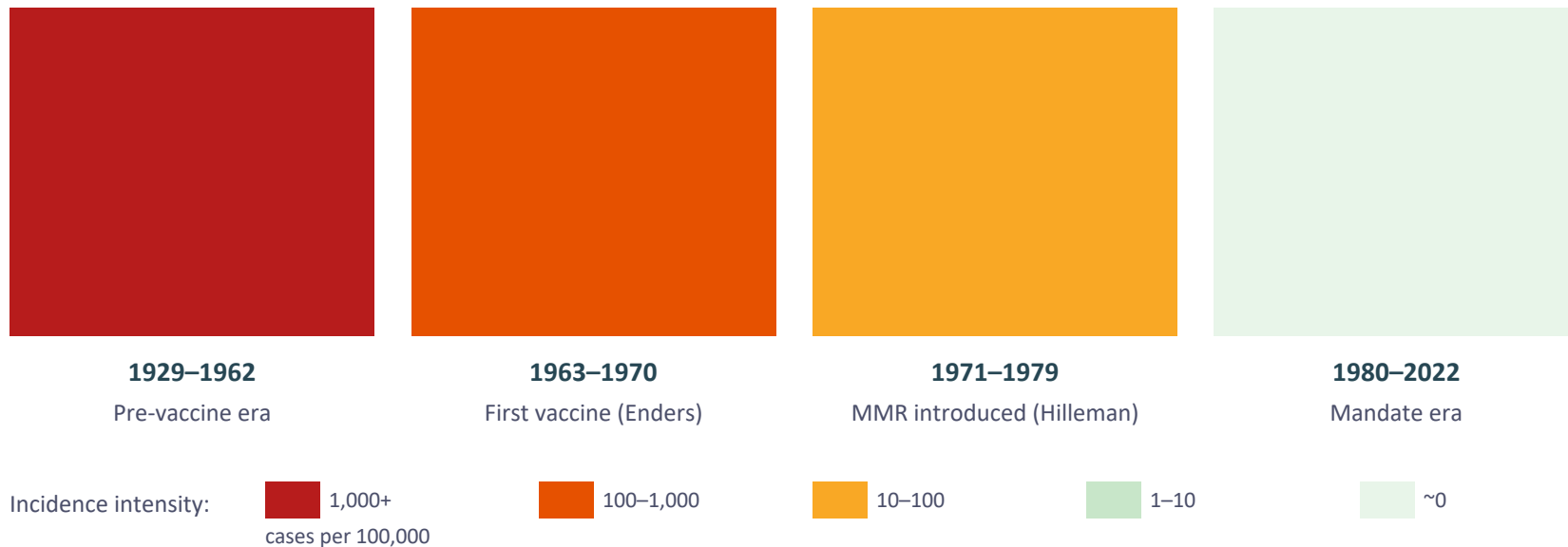
Pre-vaccine vs. post-vaccine annual rates per million in the United States

Disease	Pre-Vaccine Cases/M/yr	Post-Vaccine Cases/M/yr	Case Reduction	Pre-Vaccine Deaths/M/yr	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Smallpox	250	0	100%	2.9	100%
Acute Polio	141	0	100%	10	100%
Measles	3,044	0.2	99.99%	2.5	100%
Rubella	242	0.04	99.98%	0.09	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Hepatitis A	465	51	89%	0.5	88.7%
Hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

Green cells = 100% elimination achieved | Source: Roush & Murphy, 2007

Measles Cases Collapsed Across All 50 US States

After Vaccine Mandates



● 1963: First measles vaccine developed by John Enders

● 1971: MMR vaccine developed by Maurice Hilleman

● 1980: Vaccination became mandatory for kindergarten entry

Source: Our World in Data — Measles cases per 100,000, all 50 US states, 1929-2022

Smallpox: The Only Human Disease Eradicated Through Vaccination

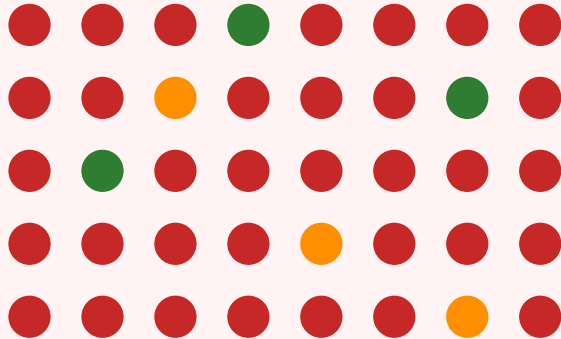


Why Smallpox Was Uniquely Suited for Eradication

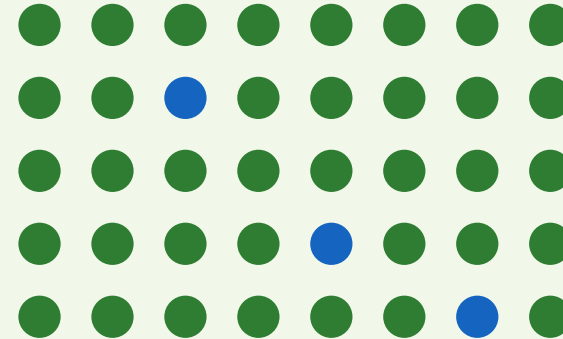
- The virus infected only humans — no animal reservoir
- Symptoms were clearly distinguishable from other diseases
- The vaccine was cheap and highly effective

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

Low Vaccination Rate



High Vaccination Rate (Herd Immunity)



● Vaccinated

● Unvaccinated & infected

● Vulnerable (cannot be vaccinated)

● Vulnerable but protected by herd immunity

Vaccines protect newborns, immunocompromised individuals, and cancer patients on chemotherapy who cannot be vaccinated.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Pneumococcal

40.5%

case reduction

31.3% death reduction

Hepatitis A

89%

case reduction

88.7% death reduction

Hepatitis B

83.9%

case reduction

83.6% death reduction

Key Barriers to Full Coverage

● Access Gaps

Globally, not everyone has access to vaccines
— supply shortages put children at risk

● Trust Challenges

Poor scientific understanding can reduce trust
and uptake in communities

● Incomplete Protection

Some vaccines provide partial reduction; newer
vaccines and better coverage strategies are
needed

Key Takeaways

Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- 01** Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and polio in the US.
- 02** Vaccine development timelines collapsed from over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology.
- 03** Measles cases dropped to near zero across all 50 states following the 1980 kindergarten vaccination mandate.
- 04** Smallpox remains the only human disease eradicated through vaccination, certified in 1980.
- 05** Gaps persist for pneumococcal disease (40.5%), hepatitis A (89%), and hepatitis B (83.9%), demanding improved access and coverage.

Global health systems must close coverage gaps, invest in next-generation vaccines, and build public trust to finish what vaccination started.